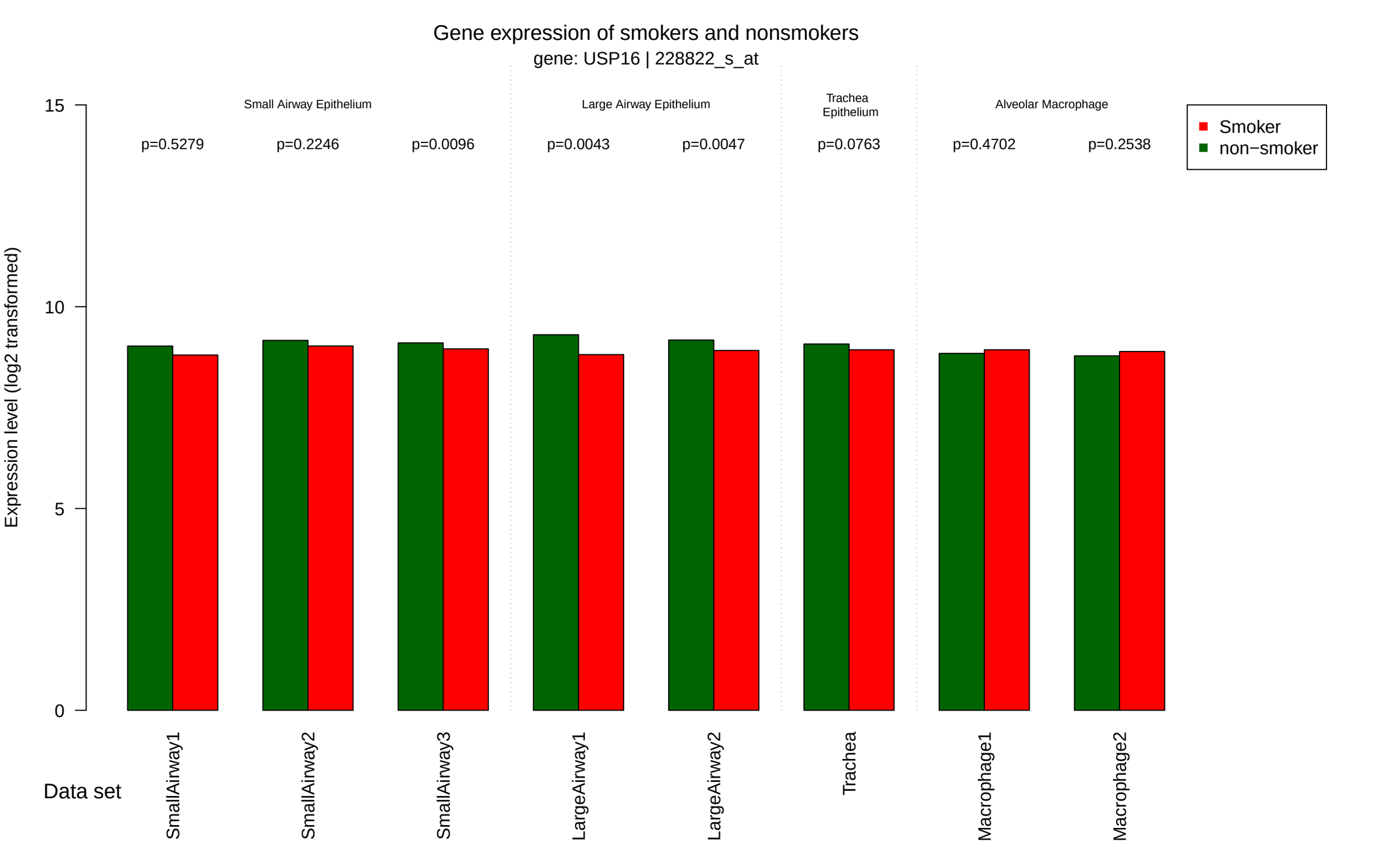
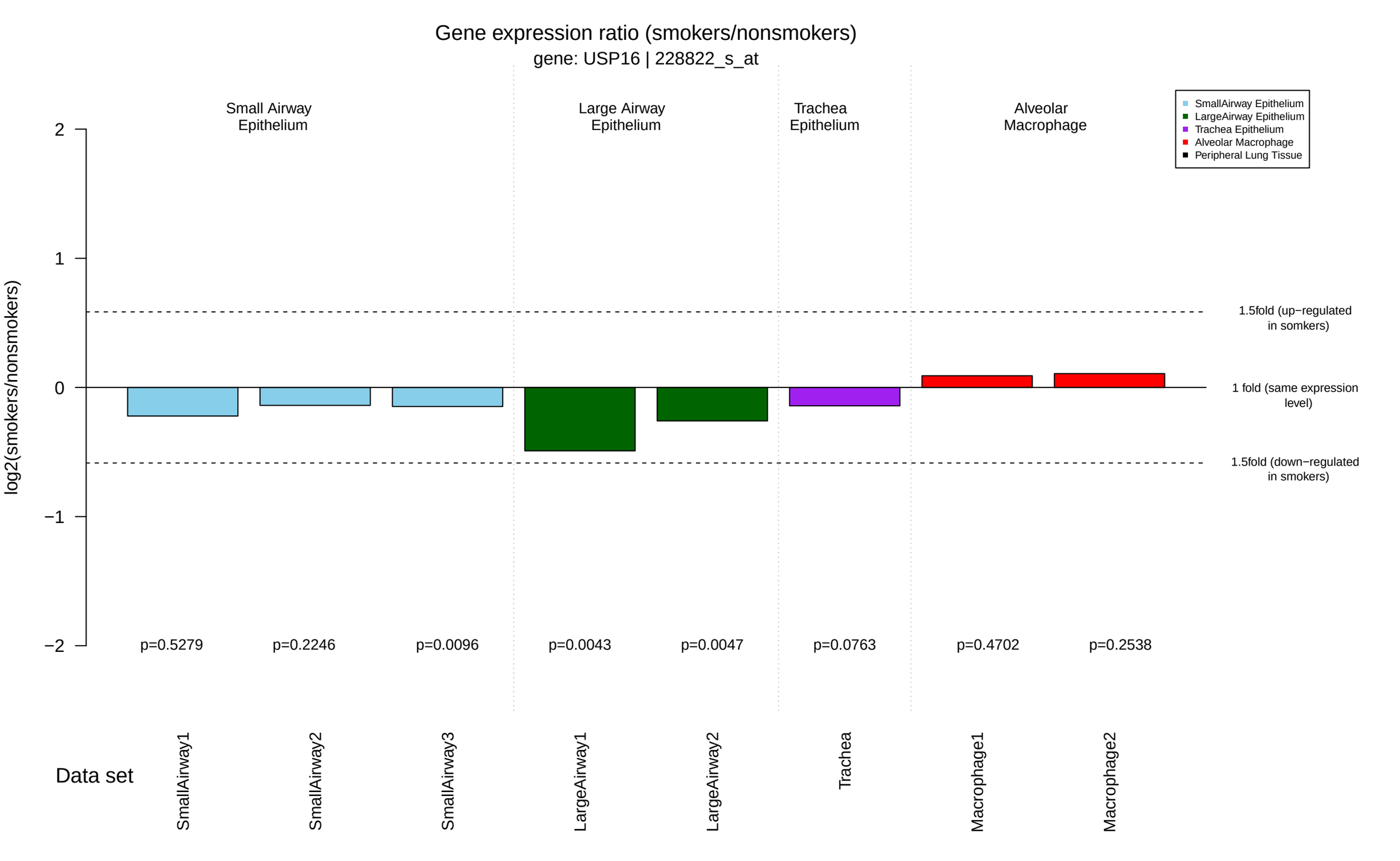
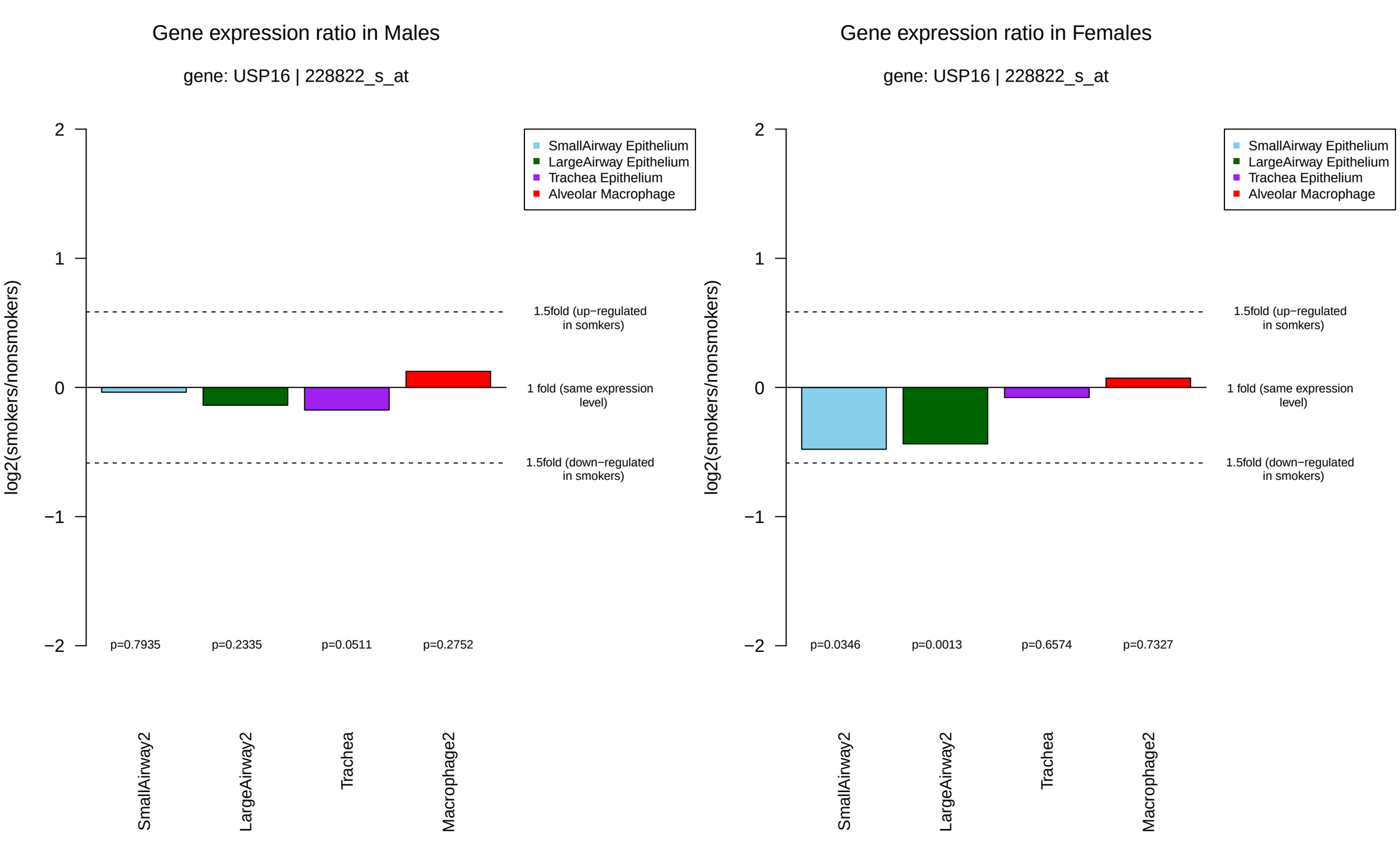
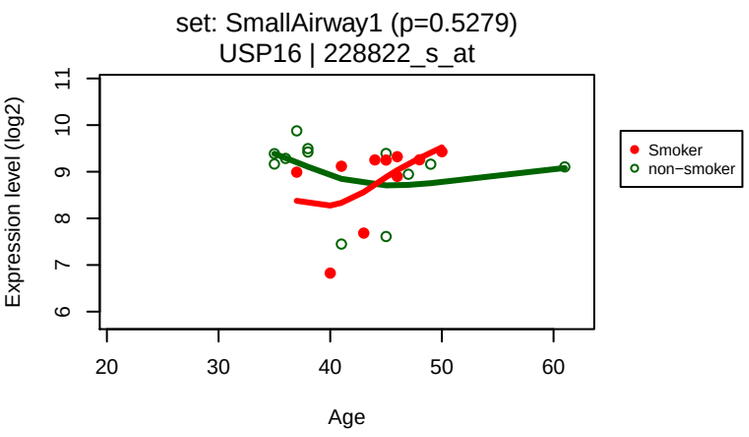
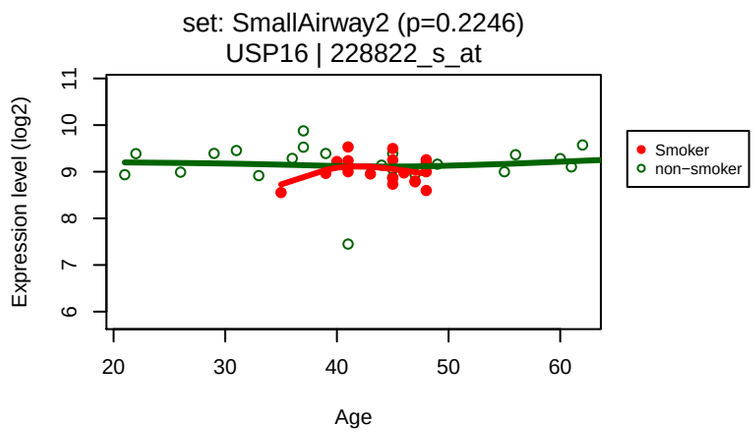
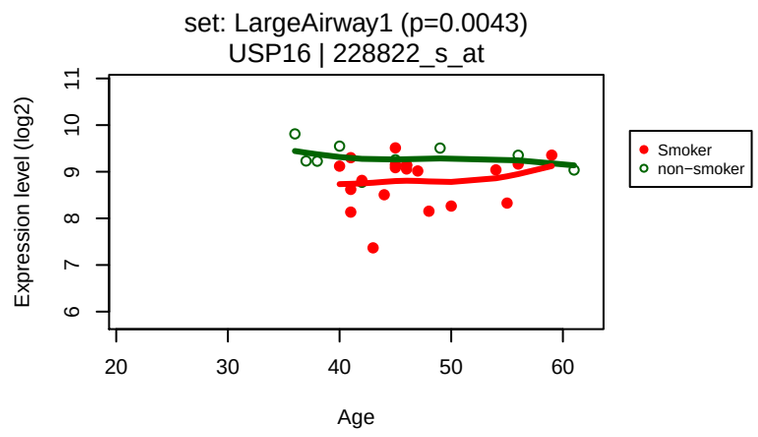

Gene expression of smokers and nonsmokers
gene: USP16 | 228822_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5279
p=0.2246
p=0.0096
p=0.0043
p=0.0047
p=0.0763
p=0.4702
p=0.2538
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

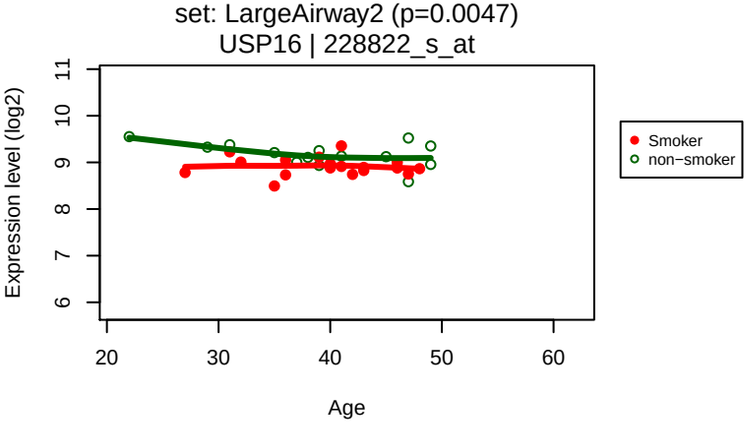

Gene expression ratio (smokers/nonsmokers)
gene: USP16 | 228822_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5279
p=0.2246
p=0.0096
p=0.0043
p=0.0047
p=0.0763
p=0.4702
p=0.2538
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

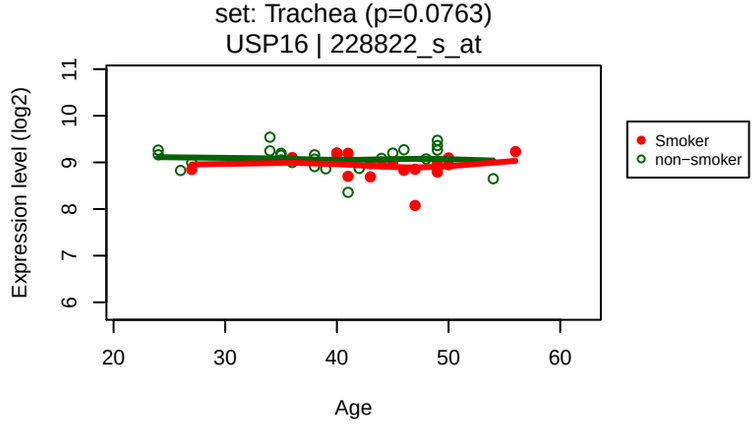

Gene expression ratio in Males
gene: USP16 | 228822_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7935
p=0.2335
p=0.0511
p=0.2752
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP16 | 228822_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0346
p=0.0013
p=0.6574
p=0.7327
SmallAirway2
LargeAirway2
Trachea
Macrophage2

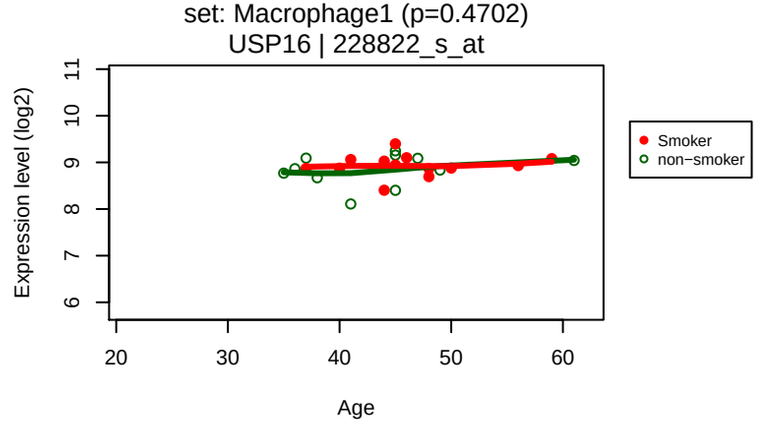

set: SmallAirway1 (p=0.5279)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker

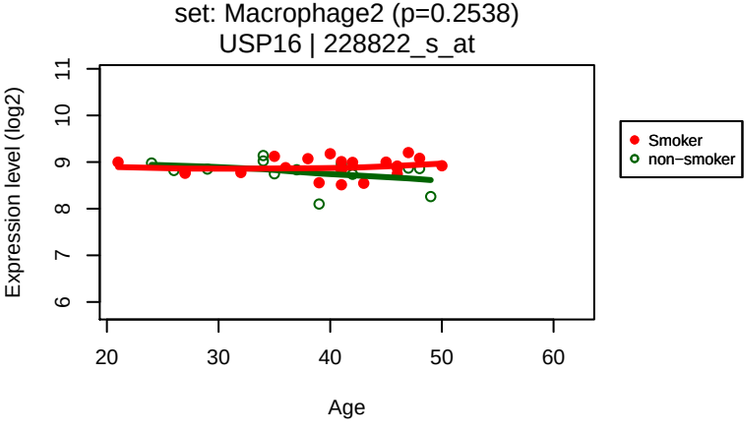

set: SmallAirway2 (p=0.2246)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker

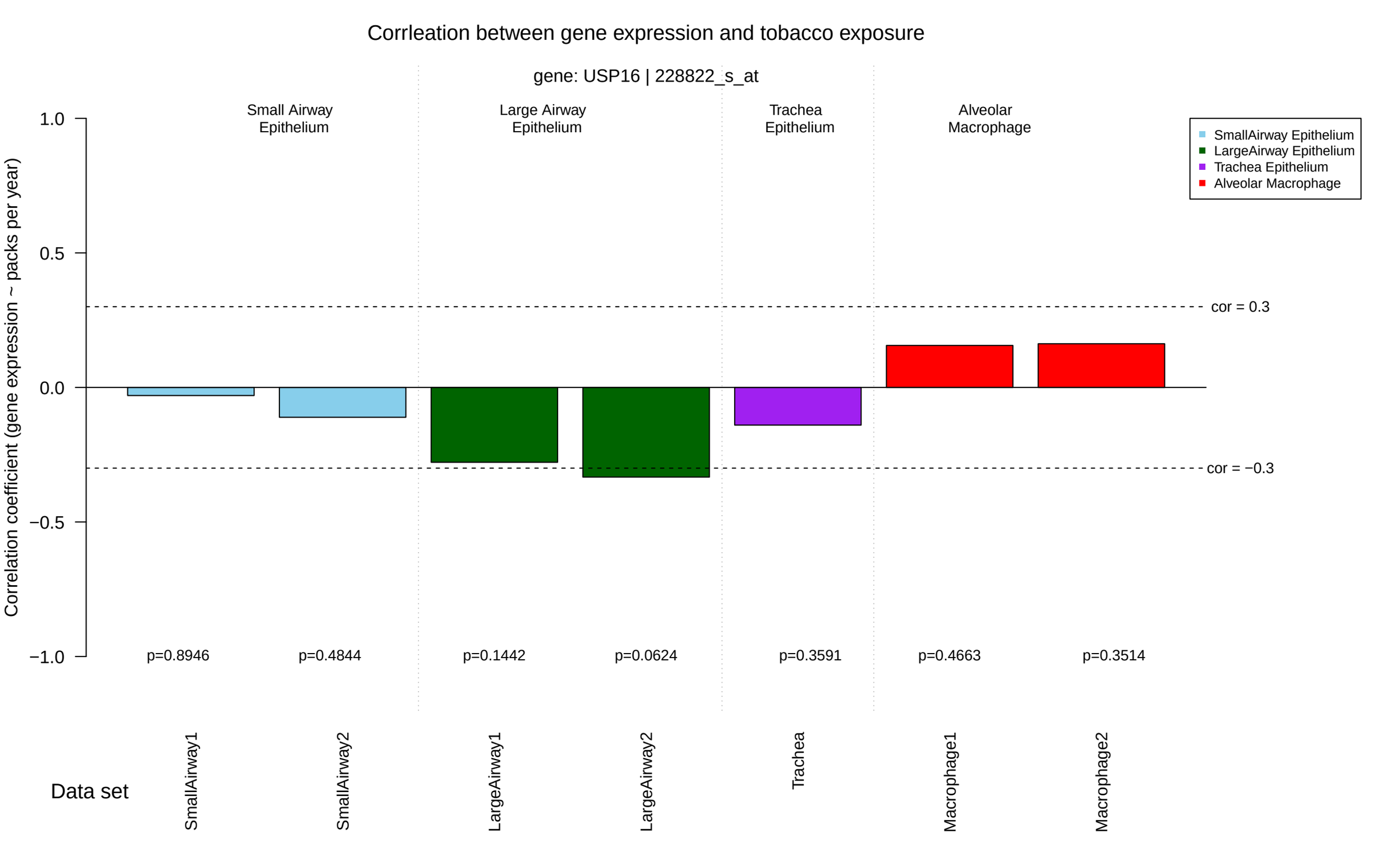

set: LargeAirway1 (p=0.0043)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0047)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0763)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4702)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2538)
USP16 | 228822_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP16 | 228822_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8946
p=0.4844
p=0.1442
p=0.0624
p=0.3591
p=0.4663
p=0.3514
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2